

Project number: CS29

Project Client: Wei Bao <wei.bao@sydney.edu.au>

Project Title: Online algorithm design for generalised ski-rental problems

Project Description and Scope:

The classical ski-rental problem [1] describes the dilemma where a customer trades off between two payment options for one commodity without the knowledge of the future: 1) renting, the customer only pays for an on-demand usage; 2) buying: the customer pays for lifetime usage at once, and no further payment will be incurred after that. To solve this problem, an online algorithm (which can process its input piece-by-piece without knowing the future) is designed. Then, we can analyse the ratio between its performance and the best offline algorithm's performance (which is called competitive ratio).

In this project, we target to study a more complicated but more realistic scenario called the two-level ski-rental problem when a decision-maker need to trade off between rental and purchase of multiple commodities (or services). For each commodity, the decision-maker can rent it or purchase it (which is called single purchase). In addition, the decision-maker can purchase all instances as a combo purchase at a reduced cost. This brings new challenges to decision-makers as payment options are now correlated, and cannot be addressed by existing solutions.

[1] Karlin; Kenyon; and Randall. 2003. Dynamic TCP Acknowledgment and Other Stories about $e/(e-1)$. *Algorithmica* 36(3): 209–224.

Expected outcomes/deliverables:

Online algorithms and competitive analysis for the two-level ski-rental problem (or a slightly modified problem).

Specific required knowledge, skills, and/or technology:

Algorithm (e.g., COMP3027/COMP9007 or equivalent); Python; knowledge in competitive analysis.

Resources provided by the client:

Related general fields/disciplines:

Algorithms; Cloud Computing; Artificial Intelligence; Data Science/Analytics;